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Introduction

For my Casual Games module I have decided to recreate the multiplayer classic Pong in such a manner that is satisfying to play on a mobile device and incorporates a scalable framework for monetization and player customization.

The game utilizes touchscreen functionality in its movement to create tactile feedback and an engagement method dependent on timing and positional awareness.

A critical social element is that the wins and losses of a given pair of unique players is permanently tracked and displayed, motivating ongoing competitiveness and returning play.

This document will discuss a build of this project, named “Pong Rivals”, at its Minimum Viable Product stage of development and scope.

Project Scope

The primary MVP scope of this project is a real time multiplayer capable build of Pong allowing two players to detect one another in a listen server dynamic, play any amount of matches with one another and have the results and consequences of those matches recorded in such a manner that the information produced by them persists to their accounts across sessions. Additionally, the MVP is built to showcase scalable systems for purchasing items that may be linked to a players account, and creating persistent friend connections with other users.

Core Gameplay Loop

The fundamental core of the gameplay loop may be summarized as the following process. To begin, the host player chooses a friend to successfully invite, a process supported by UGS and relay join code. Once the invited player accepts the invitation, they are each placed in a match and given movement control over their panel, the host controlling the leftmost panel and the invited player controlling the rightmost.

After a count down, the ball spawns in a random direction. Both players proceed to exchange deflections while failures of deflection are counted and visually displayed for each player. The first player to reach a score of three is the victor. A persistent win/loss counter linked to the relationship between two players on each other's friends list is adjusted when the match is won. Both players are then given the opportunity to vote on a rematch, if both vote to do so another match begins and if either one leaves the other player leaves the session also and the connection is terminated.

Player Action and Feedback

When a player moves their paddle, they do so on a vertical axis slider, the bar of which is logically linked to the position of the paddle, that is affected by drag physics that slow down or speed how fast your paddle moves. This results in sharp reversals of direction the paddle is currently moving being stiffer and slower relative to the speed it is moving and movements that follow current momentum of the paddle being faster and smoother. The speed a paddle is moving on collision with the ball affects the speed the ball moves away from it, likewise the angle at which both objects collide with each other affects the specific direction the ball is sent back in.

At the end of a normal match the players should both be rewarded with 100 virtual currency given to the victor, and 50 being given to the loser so as to not have the match feel like a waste of time. In an extended match this should be 300 and 150 points respectively. Instead of only the winner being given a currency reward the primary incentive of play should be the match context system and the permanent win/loss counter's balance.

Session Design

Match Length and Session Flow

A typical match in the MVP lasts no longer approximately two to three minutes. This game intentionally has no match timer.

At the end of a normal match, each player may leave - the host arriving back at their list of friends and the invitee to the main menu.

They also can choose to begin a new match, this match is considered its own entity for the purposes of recorded data persistence that is logged by their win/loss counter in its own right any number of times. Rematches may be prompted by either player but must be confirmed by both.

Interruptions, Disconnection and Abandonment

In the MVP prototype, an intentional leave by either player results in a match termination in which neither player is counted as the victor in persistent data storage. Regardless of who leaves, the termination is synchronized by the host (who is acting as a server entity as well as a client in their own right) and their side is the one that ends the Relay connection/UGS session. In the prototype, there is no functionality for players to resume a match interrupted by accidental leaving, phone calls, network loss and the like.

Customisation System

The primary variation between sessions is in aesthetic potential, the system is designed to support visual customization of the paddle boards, match border, ball, background and effects for deflection and missed shots. However, for the MVP build, this cosmetic system only applies to visual changes of the match borders, backgrounds and paddles, with more advanced visual changes not being implemented but the underlying, scalable support system for them existing in a such a way that these additions could be non obtrusively added to the existing framework during further development.

All of the aforementioned may be purchased with virtual currency earned from time investment. In the prototype build, cosmetics exist on a local basis and therefore each player may see their customized visual assets independent of what the other player sees on their end. These items are integrated into a system where each may be

customized to unlock only after a certain number of total matches or wins by a player. Unlocked items, owned items, applied items and possessed currency are all features enabled by the games persistent data management, and therefore maintain themselves on an account basis between sessions as they are changed and affected. Hosts may also enter in the social context the competition exists to settle (picking a restaurant, film to watch, petty argument etc.).

Accessibility

This game is designed with unobtrusive ease of use and rapid onboarding as priorities and necessities.

The intent behind the game is good natured competitiveness and a fun way to settle petty real life disputes, meaning that any difficulty with session setup or gameplay complexity would be terminal to its appeal.

A single axis directional choice being the singular gameplay element the player must concern themselves with results the main gameplay loop being intuitive and self explanatory, while in the menu adding friends and starting ongoing sessions is designed to be easily achievable without reasonable room for error.

Information display is also largely minimal and obvious in its intent, all that is communicated to the player is the displayed reason for the match, the win/loss counter, each player's score and their individual directional slider.

All controls are based on finger based tap and drag mobile control and so do not need tooltips or meta explanation to be understood by the player.

The game loop is also intended to be quick and unobtrusive in each player's life as matches last very short amounts of time.

Engagement Mechanics

Core Retention and Rivalry

Mechanics in this game geared towards attracting and retaining user engagement are multi-layered. At its most fundamental level, the unobtrusive and rapid loop of creating a match, inviting a friend and progressing through the match can be considered to be a positive strategy for encouraging repeat user engagement in of itself, as can the core gameplays simple and intuitive controls. These elements enable ease of spread from an initial player to their friends list due to the system having such an ease of onboarding and invite.

The most critical feature for user retainment and repeat play is the existence of a win/loss counter specifically tied to the sessions of a unique user pair as this motivates the continuance of a relationship between users who know each other outside of the games context.

Additionally, an extension of the system is its application to persistent and defined groups of players wherein each player may have their wins and losses against the other players in their group recorded and displayed, motivating competitiveness on a larger scale than one versus one.

Our approach to reward structures and in game economy also encourages committed user engagement and the introduction of new users to the game as currency for cosmetic purchases is awarded through the expansion of your friend list and repeated sessions with said friends.

Gifting

This function is out of scope for this project's MVP but the ability to purchase cosmetic items for other players would serve as an extra enticement feature for repeated play by users who weren't the initial adoptee.

Mini-Games

This concept is neither needed nor desirable for a system of this level of simplicity. The core Pong match already functions as a short, self-contained, casual activity.

Session Management and Return Rewards

As unique sessions are intentionally tracked and measured, it becomes possible to provide committed and returning players with rewards and incentives based on their progress past decided milestones of sessions or wins (e.g. 50, 100, etc.). These rewards may range from greater currency payouts, to free cosmetic items, or unique items and effects that cannot be purchased with the normal currency. As this principle applies to any unique player pair, the incentive effect cascades across any pair established.

Randomized / Lotto Rewards

Our system does not utilize any randomized reward systems as these are categorically dissimilar from our emphasis on repetitive casual play. The reward systems present are based around the concepts of encouraging routine and cascading connection between users in engagement and invite, rather than exploiting a fear of missing out or any other kind of timed pressure system.

Progression Design

Progression Overview

In this game system progression is achieved through time investment and inviting new players, a skill based progression system would be detrimental due to the simplicity of the core gameplay.

Progression materializes in terms of a player's currency gain, unlocked cosmetics, social statistics with their connected friends and overall win/loss record.

Time investment (through the match counters) is also rewarded with in-game currency that can be used to purchase variations of the aforementioned environments, cosmetics and animation changes.

Soft Currency

There is one currency system in the MVP. It is awarded to each player on the completion of a match, with the winner receiving 100 and the loser receiving 50. This currency is used to purchase cosmetic items that are unlocked by cumulation of wins and total matches played. There is no functionality of this currency or any other type of progression to alter your gameplay or ability.

Unlocks and Cosmetic Progression

As the default skins of items are understood as structurally equivalent to optional, purchaseable ones they are listed in the store and are unlocked by default to avoid a player having no visible texture for any of the elements in their session.

The cosmetics system is built around the capacity for certain items to be marked to be unlocked upon a player's tracked and recorded data hitting certain numbers. This means that customization is both fueled by and encourages user retention while also existing in the context of in-game economy.

Pair-Specific Progression

User relationships in this system exist between two authenticated UGS player IDs. These relationships are structured in such a way that they will always be mutual and a connection between a first player to a second player cannot exist without its inverse. These relationships form the basis for tracking data recording each player's wins against the other, as well as total matches played between the two distinct users.

Monetisation Strategy

Monetisation Overview

Due to its nature this section is entirely discussing a more advanced, commercial release version of Pong Rivals. In this full release hypothetical, this game system would be primarily reliant on advertisement based monetization to turn a profit with auxiliary dependency on purchases of in-game currency with real money to expedite the process of unlocking the progression rewards described above. It would also be possible for players to watch an advertisement to completion or watch more advertisements to gain in-game currency, a system beneficial to both developers and players as it generates the former profit and would to the latter be an essentially free reward. A way in which this system would be made to further profit off of what players would perceive as a natural and unobtrusive process that they would be doing to some extent anyways, is to give generous currency rewards to players for successfully adding new friends to their list (with, if possible, an increase if they are a new player altogether) as bringing new players to the game or increasing the amount of matches existing players play necessarily increases the amount of advertisements watched overall and so the developers profit. Additionally, to encourage more sessions with more users on a players friends list for the same reason as before, currency rewards are tied to session milestones as they relate to unique player pairs as well as a players overall statistics,

meaning that playing the same small amount of repetitions with several users each is as, if not more beneficial than if milestone rewards only occurred at increasing session counts across all unique player pairs.

Virtual Economy

The economic support system of this application, which would be free in a fully released state, is designed around the principle that most unlockable progression items should be technically earnable without spending money so long as a player is a committed user who is willing to regularly play and engage with their incentive to invite new players. These earnable items and effects would, in a developed build, also be purchasable should a user not want to wait for them to unlock from hitting session milestones. Alongside these purchasables, there would exist items and effects of particular desirability that would only be purchasable by real money to categorically separate them from standard items.

Real-Money Purchases

This system is monetized around a skin and effect economy model. While a full release build may include the purchases that may not be attained by freely achievable in-game currency, there is no element of the game loop that may be altered or improved by a real world money purchase.

Fairness and Player Choice

Due to the established progression fundamentals of the game, any hypothetical monetization would be entirely optional and centered around speeding up progression rather than altering the essential systems and experience of the game loop.

Social Networking Features

Social Design Overview

As this game intends to have mild real world application and which is dependent on existing context between its users for its simplistic gameplay to have lasting appeal or effect, implicit social elements are critical to its design.

The most obvious and fundamental of these being that competitive scores are tracked on user to user basis, resulting in fundamental perpetual competition as this score affects aesthetic presentation and is displayed prominently during a match between those two players.

To further facilitate the real world context in which this game would apply, a host user may specify the reason for the current match which would be displayed in their invite to the challenged user and during gameplay. The competitive nature of the game paired with its simplicity is also designed in such a way to prompt good natured “feuds” between players who know each other in person similar to Facebook’s “poke” feature, as there will always be one player ahead of the other and it always be rather achievable for the opposing player to regain the upper hand due to the game being having a low skill ceiling based in reactivity and physics awareness.

Friends List and Social Graph

To represent the systems intentions of a persistent link between two wholly separate user accounts, the prototype simulates such a connection using two distinct UGS account entities that may exist in the same build on the same device whilst having the same relationship. These entities are built off of a local simulation a demo friend account exists in to create relationship data with the user.

Social Plugins or Services

Outside of the backend and multiplayer services provided by UGS, the MVP does not make use of any integration with another social system or platform.

Challenge and Invitation Interaction

In a higher build of the game, the intended challenge structure works based on invitations sent from one entirely independent account to another that may be accepted or rejected. The challenge interface within the MVP utilizes UGS services and relay join code to create a simulation of this between two accounts running distinct from one another in the same build.

Virality and Replayability

Replay Motivation

This game's replayability and capability for viral spread is dependent on it being used to settle disputes of a funny nature and create entertaining elements of stories by its use in context. As mentioned previously the inherent competitiveness of the tracked win/loss ratios result in implicit motivation for two players to continue playing matches against one another.

Friend Invitations

This concept has no relevance to the created MVP and thus may only be discussed in the context of a full release. The invitation link feature for this system is connected to a phone's normal function. Inviting a new player sends an auto-formatted text to them that contains a link which will seamlessly begin installing the app which will automatically open upon install completion and finish the process of adding each player to each other's friends list.

Referral Incentives

While also not relevant to the prototype as it exists for the purposes of this assignment, an advanced feature discussed in concept is currency and cosmetic rewards being paid out to a user upon the successful inviting of a friend to make an account.

Player Engagement Features

Notification System

This is another conceptual feature of little relevance to the existing prototype but worthy of conceptual discussion in the context of a higher build. Adding a player with an existing account (an “account” simply being a registered google play ID) to one's friends list would be intended to be immediate and not require reciprocation. Inviting an existing player on your friends list to a match would take advantage of a phone's in-built notification system, the notification an invitee receives containing a link that will start the app and load them into a session with the user who sent the invite.

Return Motivation

Player retention has is and would be achieved in this system by the ease of the game loop onboarding, the smoothness of engaging with such with a defined friend, and the persistence of your record with each friend in your list between sessions with them.

Multiplayer Design Choices

Multiplayer Design Overview

Our multiplayer design setup involves one layer at MVP stage, functionality for saved competitions between individuals on each other's friends list, and two layers at a release stage that would include functionality for groups of players with intersecting friends lists.

As discussed, I intend to implement social features as implicit and fundamental design elements, with the win/loss counter placing inherent pressure on each player to continue their recorded matches for extended periods of time along with other systems that fulfil the same general purpose and players being incentivised to introduce new users to the game with in-game currency rewards.

Included Multiplayer Features

- The ability to invite friends using a phone number from a user's contacts list
- The storage of these friends in an easily accessible list that also serves to record and display total sessions and the win/loss counter of the unique player pair
- The ability for each player in a match to prompt and vote on rematches or match-extensions
- The ability for the host to enter in a meta reason for the match to take place

Excluded Multiplayer Features

- The ability to sort these added friends into groups of more than two which have their cumulative total sessions recorded along with a ranking of each player (out of scope for MVP)
- Gifting of cosmetic features (out of scope for unreleased build)
- Cosmetic features that only appear on a single client basis, or change from one players selection to the other depending on in-game actions and conditions (out of scope for MVP)
- Invite texts and automatic app installs (out of scope for MVP)
- Voice chat (out of scope for unreleased build)

Multiplayer Architecture

This project uses a listen server variant of the client-server topology archetype. This means that in both a prototype where each account exists on the same machine and in the same build as well as also in a higher build, the host of the relay connection exists as a server entity the invited client connects to as well as being an individual client in their own right. As the authoritative partner in the connection, the host determines much of the active systems and the information shared between them and the other player they are connected to, especially those related to the position of mutual seen objects and any element related to numeracy or time specifics.

This architecture is dependent on unities Netcode For Game Objects framework, which provides concepts such as NetworkTransforms which govern shared perception of moving objects, NetworkVariables which allow for transmission of information like decided victors/losers and decisions such as rematch consent between all parties involved and thus allow the system to respond accordingly, and RPCs which more

directly focus on actions and decisions taken by individual players (such as deciding to prematurely leave a match) and applying the consequences of those actions to the shared simulation.

Backend Development

Backend support and data persistence in this project is handled through the use of the Unity Game Services framework. This framework provides services such as authentication which registers and documents distinct player identification, cloud save which maintains any kind of status associated with that identity, and cloud code which translates live occurrence into that persistent data record, relevant when a match is won or lost.

Use of AI Tools

Tool:

ChatGPT Plus (OpenAI)

Permitted development uses during this project:

- analysis of the supplied assignment/lecture resources
- decomposition of implementation into development milestones
- generation/review of C# implementation approaches
- debugging Unity/NGO compilation and runtime problems
- assistance configuring NGO, Unity Transport, Multiplayer Play Mode
- assistance integrating UGS Authentication, Multiplayer Services Sessions and Relay
- assistance integrating Cloud Save
- assistance designing/debugging Cloud Code settlement
- interpretation of Cloud Code server errors including Bad Request, JSON conversion and write-lock conflict errors
- review of networking authority and synchronization choices

AI-generated material was used as development assistance and was reviewed and adapted during implementation according to needs and best practice.